INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

End Semester Examination, December - 2024

Degree: **B.C.A**, Semester: **First**

Stream: **BCA**

Paper Code: **BCACC101**

Paper Name: **Introduction to C Programming**

Full Marks:100 Duration:3 Hours

Part - A

Attempt 10 questions out of 15 questions

Each question carries 2 Marks (2 X 10)

1. Find errors, if any, in the following program: **2**

```
/* A simple program
int main ( )
{
        /* Does nothing */
}
```

2. Why and when do we use the **#include** directive? **2**

3. Describe the structure of a C program. **2**

4. Write a program to output the following multiplication table: **2**

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
  .   .
  .   .
5 x 10 = 50
```

5. Why and when do we use the #define directive? **2**

6. State the differences between the declaration of a variable and the definition of a symbolic name. 2

7. Write a program that prints the even numbers from 1 to 100. 2

8. Find the output of the following program? 2

```
main ( )
{
        int x = 100;
        printf ("%d/n", 10 + x++);
        printf ("%d/n", 10 + ++x);
}
```

9. What is the output of the following program? 2

```
main ( )
{
        unsigned x = 1;
        signed char y = -1;
        if (x > y)
                printf ("x > y");
        else
                printf ("x < = y");
}
```

10. Write a program to print the size of various data types in C. 2

11. Write a program that determines whether a given integer is odd or even and displays the number and description on the same line. 2

12. How can we use the **putchar ( )** function to output multicharacter strings? 2

13. A **switch** statement can always be replaced by a series of **if..else** statements. Explain whether the statement is correct or not. 2

14. The predicate !((x > =10) ¦ (y = =5)) is equivalent to (x < 10) && (y! =5). Explain whether the statement is correct or not. 2

15. Which statement codes the ternary conditional expression using the operator ?: ? 2

Part - B

Attempt 6 questions out of 9 questions
Each question carries 5 Marks (5 X 6)

16. Find the errors, if any, in each of the following segments: **5**

a)
```
if (x + y = z && y > 0)
        printf (" ");
```

b)
```
if (code > 1);
        a = b +c
  else
        a = 0
```

c)
```
if (p < 0) || (q < 0)
        printf ("sign is negative");
```

17. Shown below is a Floyd's triangle. **5**

```
1
2 3
4 5 6
7 8 9 10
11 .. .. .. 15
.
.
79 .. .. .. .. .. 91
```

a) Write a program to print this triangle.

b) Modify the program to produce the following form of Floyd's triangle.

```
1.
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

18. How can we use **for** loops when the number of iterations are not known? **5**

19. Write an interactive program that could read a positive integer number and decide whether the number is a prime number and display the output accordingly. **5**

Modify the program to count all the prime numbers that lie between 100 and 200.

20. The numbers in the sequence 5

1 1 2 3 5 8 13 21 .....

are called Fibonacci numbers. Write a program using a **do.... while** loop to calculate and print the first m Fibonacci numbers.

21. What happens when an array with a specified size is assigned 5

a) With values fewer than the specified size; and
b) With values more than the specified size.

22. Given are two one-dimensional arrays A and B which are sorted in ascending order. Write a program to merge them into a single sorted array C that contains every item from arrays A and B, in ascending order. 5

23. Write a function that receives a sorted array of integers and an integer value, and inserts the value in its correct place. 5

24. Explain the facilities provided by the C pre-processor with examples. 5

Part - C

Attempt 5 questions out of 8 questions

Each question carries 10 Marks (10 X 5)

25. Write a program to compute and display the sum of all integers that are divisible by 6 but not divisible by 4 and lie between 0 and 100. The program should also count and display the number of such values. **10**

26. Describe the limitations of using **getchar** and **scanf** functions for reading strings. **10**

27. Write a program to read two strings and compare them using the function strncmp ( ) and print a message that the first string is equal, less, or greater than the second one. **10**

28. Summarize the rules for passing parameters to functions by pointers. **10**

29. Write a function that will scan a character string passed as an argument and convert all lowercase characters into their uppercase equivalents. **10**

30. Write a function that can be called to compute the product of two matrices of size m by n and n by m. The main function provides the values for m and n and two matrices. **10**

31. Write a program using pointers to read in an array of integers and print its elements in reverse order. **10**

32. Enumerate the differences between functions and parameterized macros. **10**

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